

Digital Distribution Operating System (DDOS)

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YOUR COMPLETE SYSTEM (SIMPLIFIED)

Brand → Territory → Distributor → Salesman → Retailer → Order → Delivery → Payment → Accounting

5 CORE LAYERS YOU MUST DOMINATE

1 MASTER CATALOG (CONTROL)

- Super admin creates all SKUs
- Standard naming, units, weight
- Brand mapping

 Outcome:

- No duplication
 - Clean ecosystem
-

2 TERRITORY CONTROL (YOUR MOAT)

- Geo-fencing per brand + distributor
- No overlap
 - While creating the brand, add an option that determines whether it is allowed to overlap or not
 - If the same product is delivered by more than one distributor, then show each distributor's price to the retailer when adding an item to the cart.
- Retailer locked to distributor

 Outcome:

- Distributor trust
 - No price war chaos
-

3 PRICING + SCHEME ENGINE

- Slab pricing
- Group schemes -- phase 2
- Salesman discounts

👉 Outcome:

- Matches real FMCG selling behavior
-

4 🚚 EXECUTION ENGINE (YOUR DIFFERENTIATOR)

- Delivery tracking
- Partial delivery
- Cash + UPI tracking + **signature + cheque (distributor can enable/disable options)**
 - Cheque: check number and cheque image
 - Signature: sign on the invoice
- Settlement

👉 Outcome:

- Eliminates biggest pain in FMCG
-

5 📊 CONTROL & VISIBILITY

- Distributor dashboard
- Salesman performance
- Retailer activity

👉 Outcome:

- Data-driven decisions
-

🔥 WHY THIS MODEL WINS

Traditional FMCG has:

- Manual orders
 - No delivery tracking
 - Cash leakage
 - No visibility
-

You replace that with:

Traditional	Your System
-------------	-------------

WhatsApp orders	Structured ordering
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Traditional	Your System
Verbal schemes	System-driven schemes
Driver memory	App tracking
Cash confusion	Clear settlement
Relationship-based	Data-driven

YOUR REAL MOAT (IMPORTANT)

Your moat is NOT:

-  App
 -  Features
-

Your moat is:

- 1. Territory control**
 - 2. Distributor lock-in**
 - 3. Daily usage dependency**
-

WHAT YOU MUST GET RIGHT

1. DRIVER FLOW (NON-NEGOTIABLE)

If driver doesn't use → system fails

2. PAYMENT TRACKING

This builds trust

3. SIMPLE UI

Your users are not tech-savvy

OVERALL FLOW (CONTROL TOWER)

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Order Cutoff → Order Processing → Vehicle Assignment /driver -> Zone Grouping → Loading Slip → Dispatch → Delivery

1) ORDER CUTOFF LOGIC (AUTO SYSTEM)

Rule:

Before 3:00 PM → Next Day Delivery

After 3:00 PM → Day After Delivery

Backend Logic:

- System auto tags:
 - delivery_date
 - cutoff_batch
-

UI (Admin Dashboard)

Today Orders (Before 3 PM): 120

Next Batch (After 3 PM): 45

👉 No manual confusion

👉 Fully automated

2) ORDER PROCESSING (3 PM – 5 PM)

Order Processing UI

6

Screen:

Orders for Next Day Delivery

INV101 | Mahadev | ₹8500 | Zone: Sarkhej

INV102 | Patel | ₹6200 | Zone: Narol

[Approve / Modify / Hold]

👉 Distributor can:

- Edit qty (if needed)
- Hold order

📌 **3) ZONE GROUPING (VERY IMPORTANT)**

System Groups Orders By:

- Area / delivery zone

UI:

Zone: Sarkhej

Orders: 25

Total Weight: 2500 kg

Total Value: ₹2,50,000

Zone: Narol

Orders: 18

Weight: 1800 kg

👉 This is your **planning layer**

🚚 **4) VEHICLE ASSIGNMENT (SMART FLOW)**

Vehicle Assignment UI

Screen:

Select Vehicle:

Vehicle: Tata 407
Capacity: 3000 kg

Zone: Sarkhej
Weight: 2500 kg 

[Assign Driver: Ravi]

[Confirm Load]

 Rules:

- Total assigned weight $\leq$ vehicle capacity
- Show remaining capacity

 5) LOADING SELECTION (SEMI-MANUAL CONTROL)

Logic:

Distributor can:

- Select full zone
OR
- Select specific invoices

UI:

Selected Orders:

INV101 → 200 kg
INV102 → 150 kg
INV103 → 300 kg

Total: 2500 kg / 3000 kg

[Add More / Remove]

👉 Gives flexibility

📄 6) DISPATCH READY → LOADING SLIP

Loading Slip UI

6

Button:

[Generate Loading Slip]

Generated PDF:

◆ **Header:**

Loading Slip No: LS-1001
Date: 28 Apr
Driver: Ravi
Vehicle: GJ01XX1234
Zone: Sarkhej

◆ **Items Summary:**

Sugar 1kg → 500 units
Oil 1L → 200 units

◆ **Invoice List:**

INV101 | Mahadev | ₹8500
INV102 | Patel | ₹6200

👉 Used for:

- Warehouse loading
- Driver reference

🚀 **7) DISPATCH FLOW**

After slip generation:

Status → Dispatch Ready → Dispatched

👉 Locks:

- Orders
- Quantities

🔄 **8) INTEGRATION WITH DRIVER APP**

Driver sees:

- Assigned loading slip
- Route
- Orders

👉 Perfect sync

⚙️ **9) PRICING & STOCK UPDATE (DAILY)**

Distributor Panel:

Product: Sugar 1kg

Price: ₹50

Stock: 2000 units

[Update]

👉 Optional but powerful

🔒 CRITICAL RULES

1. No order editing after dispatch
 2. Weight must match load
 3. One loading slip = one vehicle
 4. Full audit trail
-

🔥 WHAT MAKES THIS SYSTEM POWERFUL

You are controlling:

- 👉 Order flow
- 👉 Logistics
- 👉 Inventory
- 👉 Execution

MVP FEATURE CHECKLIST

MVP FEATURE CHECKLIST

1. AUTH + ROLE SYSTEM

- OTP login
 - Roles:
 - Super Admin
 - Distributor
 - Salesman
 - Driver
 - Retailer
-

2. MASTER PRODUCT SYSTEM (SUPER ADMIN)

- Create product:
 - Name
 - Parent Company
 - Brand
 - SKU
 - Unit
 - Secondary unit (outer case/bag)
 - Weight
 - Assign product → distributor
-

3. DISTRIBUTOR PRODUCT CONTROL

- Activate/deactivate product
 - Set:
 - Selling price
 - MRP
 - Stock qty
-

4. GEO-FENCING + DISTRIBUTOR MAPPING

- Assign:

- Brand → distributor → area
 - Retailer mapped to the area
 - Retailer sees ONLY the assigned distributor
-

5. RETAILER APP (ORDERING)

- OTP login
 - Distributor store (ui)
 - Product listing (filtered)
 - Cart
 - Place order
 - Order history
 - Repeat order
-

6. PRICING ENGINE

- Base price
 - Quantity slab pricing
-

7. SCHEME ENGINE (MVP LEVEL) (if possible)

- Buy X Get Y
 - Group slab discount (multi-SKU like Madhur)
 - Auto apply best scheme
-

8. ORDER MANAGEMENT (DISTRIBUTOR)

- Auto cutoff:
 - Before 3 PM → next day
 - After → day after
 - Order list
 - Edit/approve
-

9. ZONE GROUPING

- Group orders by:

- Area/zone
 - Show:
 - Total weight
 - Total value
-

10. VEHICLE + DRIVER ASSIGNMENT

- Create vehicle:
 - Capacity (kg)
 - Assign:
 - Driver
 - Orders/zone
 - Weight validation
-

11. LOADING SLIP (PDF GENERATION)

- Unique slip number
 - Show:
 - Total items
 - Item-wise qty
 - Invoice list
 - Shop names
 - Total weight
-

12. DRIVER APP (CORE ENGINE)

- Route list
- Order detail:
 - Delivered qty update
- Delivery status:
 - Delivered
 - Partial
 - Not delivered
- Payment entry:

- Cash + UPI
 - OTP confirmation (in app otp)
 - Next stop flow
-

13. PAYMENT + SETTLEMENT

- Auto amount calculation
 - Daily settlement:
 - Expected vs collected
 - Lock after submission
-

14. CREDIT NOTE (AUTO)

- Trigger:
 - Short delivery
 - Generate credit note
-

15. SALESMAN APP (LIGHT VERSION)

- Retailer verification:
 - GPS
 - Shop photo
 - Assisted order (optional but useful)
-

16. INVENTORY (BASIC)

- Manual stock update
 - Stock deduction on delivery
-

17. TALLY EXPORT (VERY IMPORTANT)

- Sales invoice export
 - Credit note export
 - Payment export
-

18. BASIC DASHBOARD (DISTRIBUTOR)

- Orders summary
- Delivery summary
- Payment summary

TESTING + UAT CHECKLIST (REAL FIELD CASES)

- _____
-  **ORDERING**
 - Retailer places order before 3 PM → next day ✓
 - After 3 PM → day after ✓
 - Repeat order works ✓
- _____
-  **PRICING**
 - Slab pricing applied correctly
 - Scheme applied correctly
 - Multiple SKU scheme works
- _____
-  **DELIVERY**
 - Full delivery
 - Partial delivery
 - Shop closed case
- _____
-  **PAYMENT**
 - Cash + UPI split works
 - No mismatch allowed
 - Settlement matches
- _____
-  **LOADING**
 - Loading slip matches items
 - Weight calculation correct
 - Vehicle capacity respected
- _____

- **ACCOUNTING**

- Sales export correct
- Credit note correct
- Payment export correct

- ---

- **GEO**

- Retailer sees correct distributor
- No overlap

- ---

- **EDGE CASES**

- Internet off (driver app)
- Wrong qty entered
- Duplicate order

CORE ROLE OF SALESMAN APP

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Not the order entry machine's main focus

👉 **Retailer onboarding + relationship + order push**

SALESMAN APP — COMPLETE FLOW

Design this in 5 parts:

1. 🏠 Home
 2. 📄 Retailer Verification
 3. 🛒 Assisted Order (powerful feature)
 4. 📊 Retailer List & Tracking
 5. 🎯 Performance / Targets (phase 2)
-

1) HOME / DAILY PLAN

Salesman Dashboard

7

UI:

Good Morning, Ravi

Today's Plan:

Retailers: 15

Pending Verification: 5

[Start Day]

List:

1. Mahadev Kirana
2. Shree Store

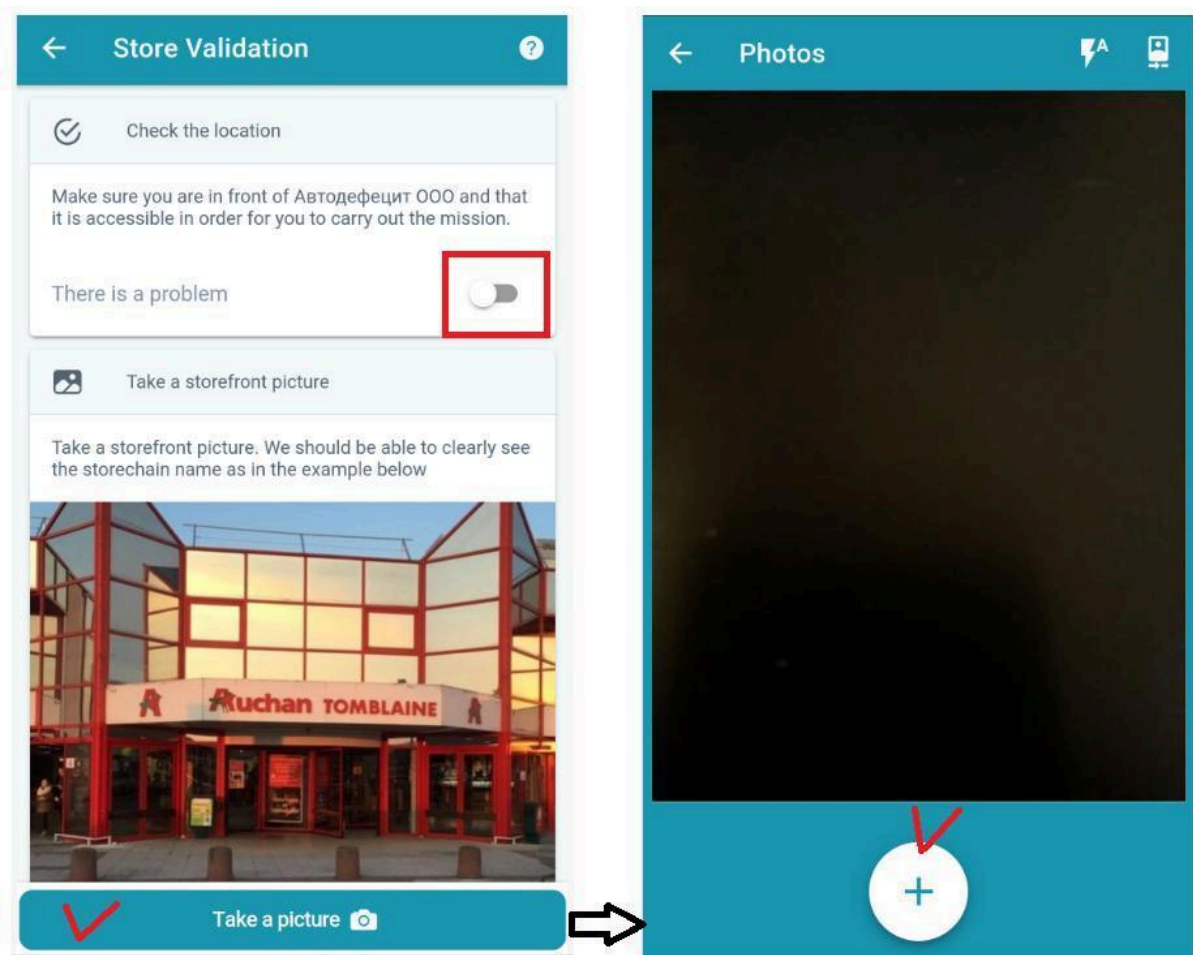
3. Patel Mart

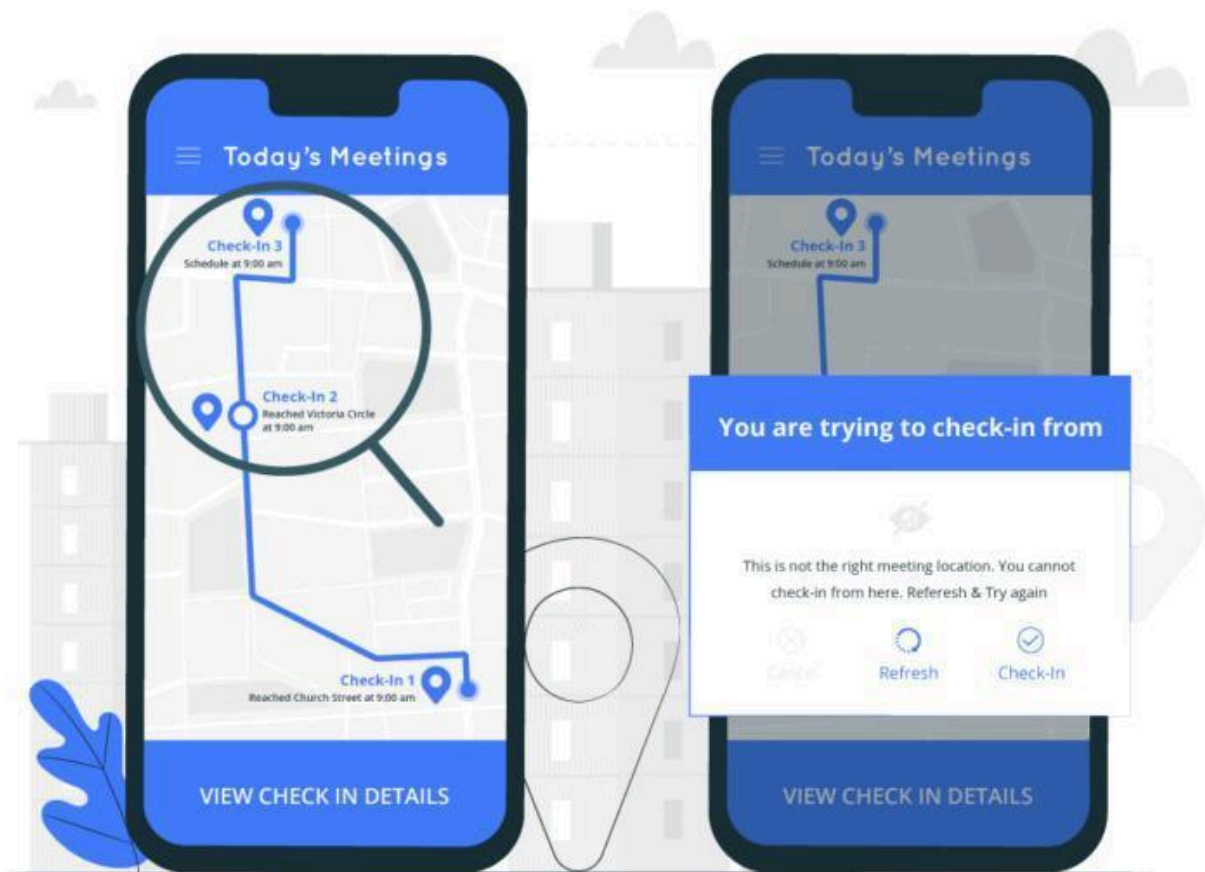
👉 Focus:

- Daily visit list
 - Pending tasks
-

2) RETAILER VERIFICATION (CORE FEATURE)

Verification UI





6

UI:

Retailer: Mahadev Kirana

[Capture Location 📍]

[Take Shop Photo 📷]

Notes:

[_____]

[Verify Retailer]

Logic:

- Capture GPS
- Capture storefront photo
- Mark "Verified"

3) ASSISTED ORDER (OPTIONAL BUT POWERFUL)

👉 Salesman helps retailer place order

Assisted Order UI

6

UI:

Retailer: Mahadev Kirana

Products:

Sugar 1kg
[- 10 +]

Oil 1L
[- 5 +]

Total: ₹8500

[Place Order]

👉 Uses same pricing engine

👉 Same cart logic

4) RETAILER LIST & STATUS

Retailer List UI

UI:

My Retailers

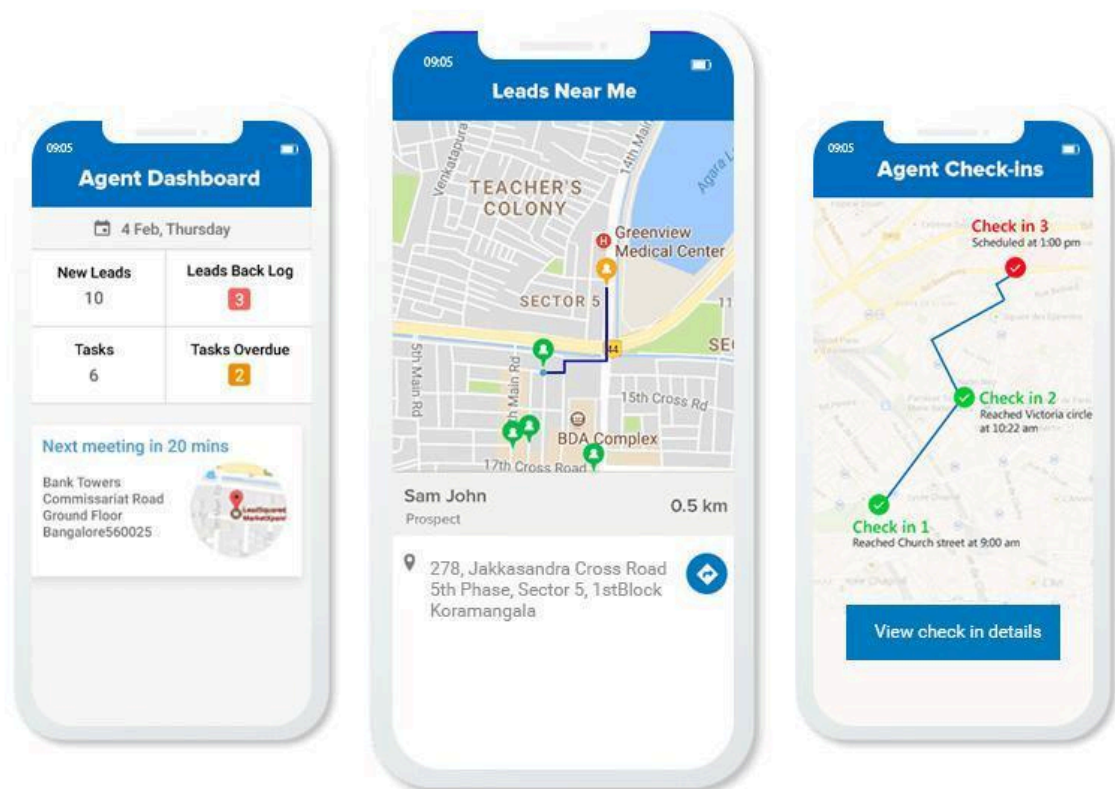
Mahadev Kirana ✓ Active
Shree Store ⌚ Pending
Patel Mart ⚠ Inactive

👉 Helps track:

- Who is active
 - Who needs follow-up
-

 5) PERFORMANCE DASHBOARD (phase 2)

Salesman Performance UI



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UI:

This Month:

Retailers Verified: 25
Orders Generated: ₹3,50,000

Target: ₹5,00,000

Progress: 70%

👉 Motivates usage

📌 COMPLETE SALESMAN FLOW

Login → Daily Plan → Visit Shop → Verify → Assist Order → Track Retailers → Repeat

IMPORTANT DESIGN RULES

1. Keep It LIGHT

- No complex screens
 - No heavy typing
-

2. Focus on FIELD USE

- Big buttons
 - Fast actions
-

3. Don't Force Usage

- Make it helpful, not mandatory

RETAILER / HORECA APP — ONBOARDING FLOW

RETAILER / HORECA APP — ONBOARDING FLOW

1) LOGIN / SIGNUP (OTP)

Login Screen

UI:

Enter Mobile Number

[9876543210]

[Send OTP]

Flow:

- Enter mobile
- OTP verify

👉 If new → go to registration
👉 If existing → login

2) REGISTRATION SCREEN

Retailer Registration UI

Fields (Keep Minimal):

Shop Name *

Party Name *

Mobile No (auto)

GST No (optional) (fetch data from gst if possible)

Address *

[Submit]

Field Meaning (Important)

- **Shop Name** → Display name

- **Party Name** → Billing name (Tally sync)
 - **GST No** → Optional (many small shops don't have)
 - **Address** → Mandatory
-

VALIDATION RULES

Mobile:

- Must be unique
-

GST:

- Optional
 - If entered → validate format
-

Address:

- Minimum length check
-

OPTIONAL (HIGH VALUE)

Auto Location Capture

Add:

- GPS capture (optional)

 Helps later:

- Delivery routing
 - Territory mapping
-

STATUS SYSTEM (VERY IMPORTANT)

After registration:

 Status = **PENDING**

Status Flow:

pending → salesman_verified → approved → active

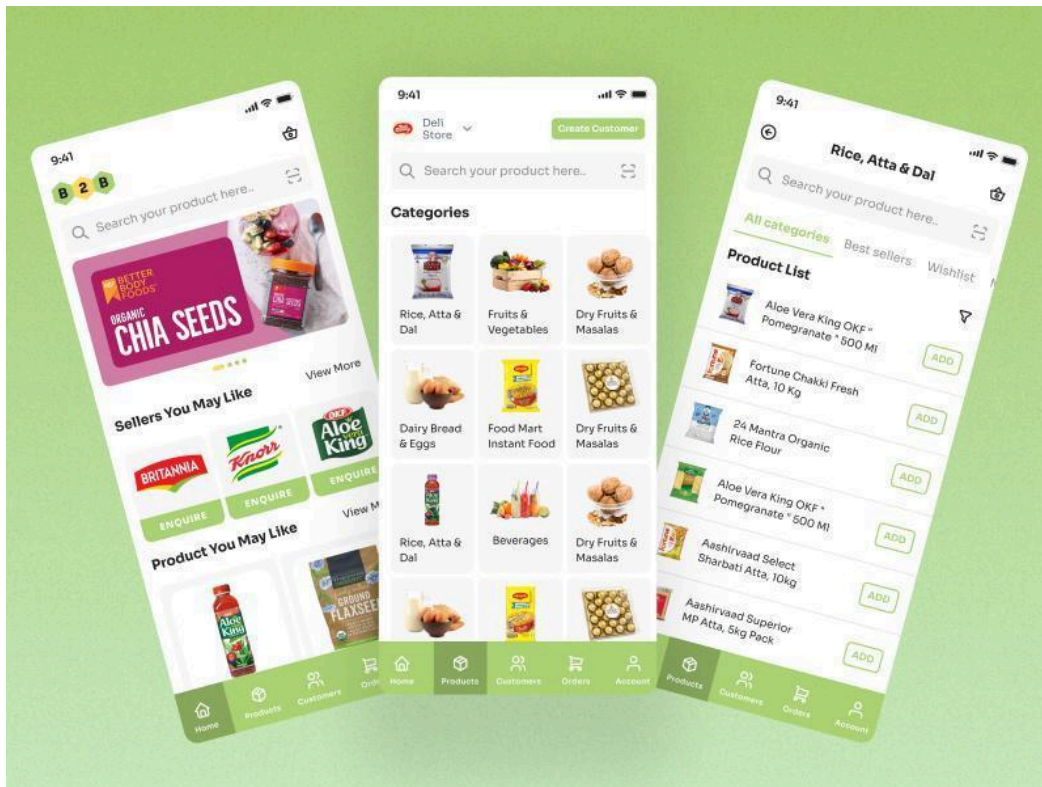
Rules:

- Pending → can browse
- Not approved → cannot order

1) RETAILER ORDERING FLOW (COMPLETE)

A) HOME / CATALOG

Product Listing UI



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UI Structure:

Search 

Category: Sugar / Oil / Snacks

Sugar 1kg

₹50
[- 1 +]

Oil 1L
₹120
[- 0 +]

👉 Show:

- Product name
 - Base price
 - Quantity stepper (multi pricing based on qty)
-
-

B) PRODUCT DETAIL (IMPORTANT FOR SLAB)

Quantity Pricing UI

UI:

Madhur Sugar 1kg * 25 pcs

Price: ₹50 each

Bag : +1250

Bulk Pricing:

1-10 → ₹50

11-50 → ₹48

50+ → ₹45

[Qty: 12]

Final Price: ₹48

👉 Key:

- Show slabs clearly
 - Update price LIVE
-

C) CART (MULTI-VENDOR SUPPORT)

Cart UI

UI Structure:

Distributor: HL Proteins

Items:

Sugar 10kg → ₹500

Oil 5L → ₹600

Subtotal: ₹1100

Minimum Order: ₹2000

Remaining: ₹900 

Distributor: ABC Foods

Items:

Snacks → ₹800

Subtotal: ₹800

Minimum Order: ₹1000

Remaining: ₹200 

 Key:

- Cart grouped by distributor
 - Show minimum order per vendor
-
-

CHECKOUT SCREEN

UI:

Distributor: HL Proteins

Subtotal: ₹1100

Minimum: ₹2000

⚠ Add ₹900 more to place order

Distributor: ABC Foods

Subtotal: ₹1200

Minimum met ✓

[Place Orders]

Checkout Behavior:

Case 1:

All vendors meet min

👉 Allow checkout

Case 2:

Some vendors fail

👉 Options:

- Allow partial checkout (recommended)

Recommended:

👉 Place order ONLY for valid vendors

🔥 FINAL FLOW

Catalog → Product → Qty → Slab Price → Cart → Vendor Group → Min Check → Order

👍 RESULT

You now have:

- Full ordering UX
- Multi pricing system

- Multi-vendor cart logic

📦 1) ORDER TRACKING SCREEN (RETAILER)

👉 Goal:

Retailer always knows “order kaha hai”

📦 Order List Screen

Order Tracking UI

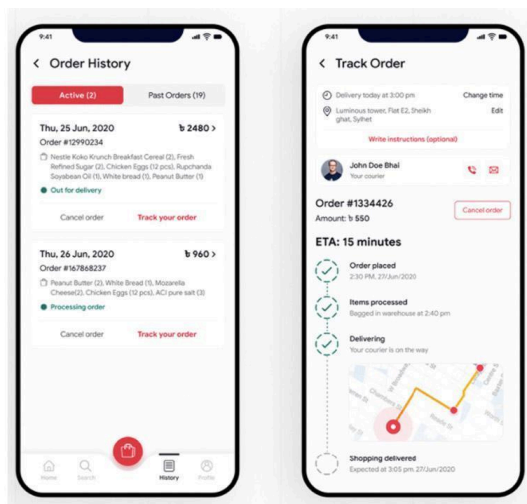
📦 1) ORDER TRACKING SCREEN (RETAILER)

👉 Goal:

Retailer always knows “order kaha hai”

📦 Order List Screen

Order Tracking UI



6

UI:

My Orders

INV123 | ₹8500

Status: Out for Delivery 🚚

INV122 | ₹6200

Status: Delivered ✓

INV121 | ₹5000

Status: Pending ⌚

Order Detail Screen

UI:

Invoice: INV123

Distributor: HL Proteins

Items:

Sugar 10kg

Oil 5L

Status Timeline:

✓ Order Placed

✓ Approved

✓ Out for Delivery

⌚ Delivered

Expected Today

👉 Keep it simple timeline

👉 No over-detail

3) RETAILER LEDGER (VERY POWERFUL)

👉 This builds TRUST

👉 Also useful if you enable credit later

Ledger Screen

Ledger UI

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UI:

Account Statement

10 Apr
Invoice INV123
+ ₹8500

10 Apr
Payment (Cash)
- ₹5000

10 Apr
Payment (UPI)
- ₹3500

Balance: ₹0

Key Points

- Show:
 - Invoice
 - Payment
- Running balance (later when credit system introduced)

👉 Even with COD → useful for tracking

COMPLETE RETAILER FLOW

Login → Browse → Add to Cart → Multi-vendor check → Order → Track → Delivery → Issue (if any) → Ledger

IMPORTANT DESIGN RULES

1. Keep it SIMPLE

Retailers are not tech-savvy

2. Avoid Over-information

- Don't show backend complexity
-

3. Focus on TRUST

- Clear order status
 - Clear payment record
-

WHAT MAKES THIS STRONG

You are giving retailer:

- Transparency ✓
- Confidence ✓
- Simplicity ✓

👉 This drives adoption

FINAL RESULT

You now have full retailer side:

- Onboarding
- Ordering

- Pricing
- Cart
- Tracking
- Issue system
- Ledger

DRIVER APP — DELIVERY EXECUTION FLOW

DRIVER APP — DELIVERY EXECUTION FLOW

Goal:

Fast, accurate, no confusion, no cash mismatch

FULL FLOW OVERVIEW

Login → Start Trip → Route → Delivery → Payment → Completion → Next Stop → End Day Settlement

1) LOGIN & START DAY

Screen:

Good Morning, Ravi

Assigned Orders: 15

[Start Trip]

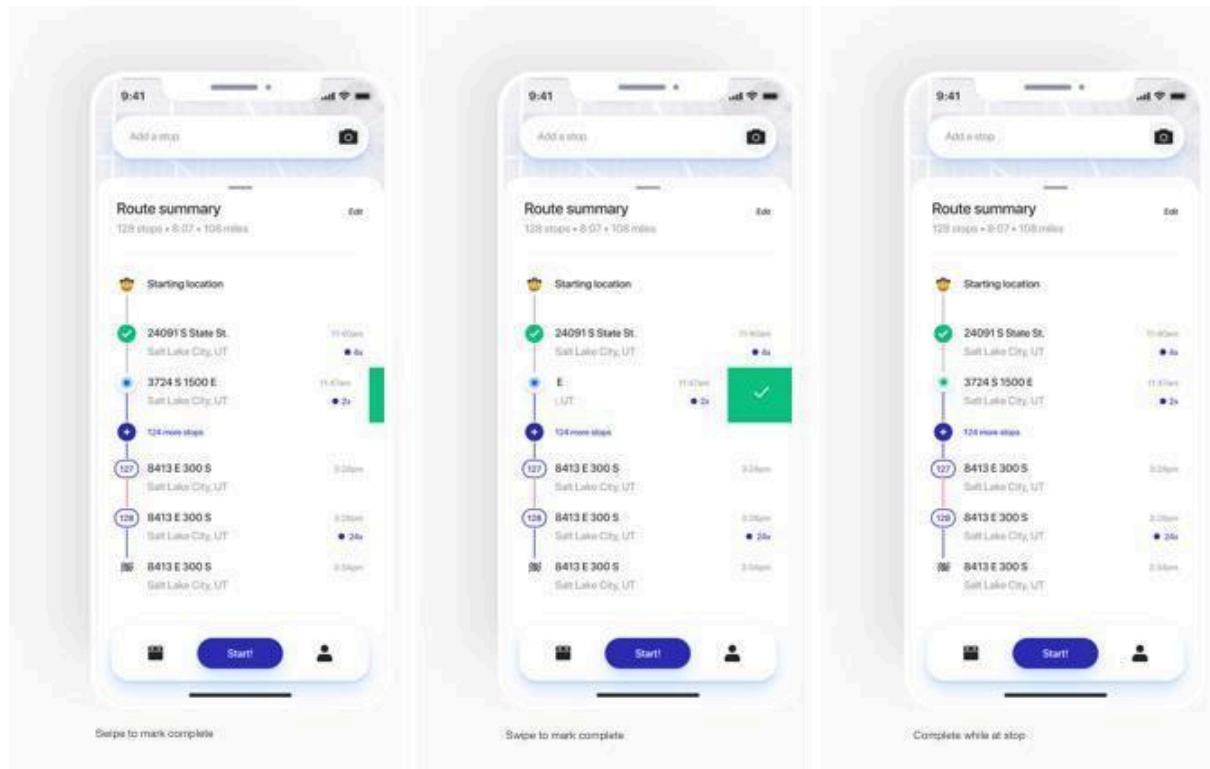
Logic:

- Driver sees:
 - Total stops
 - Expected cash

 Sets mental clarity

2) ROUTE / STOP LIST

Route UI



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Today's Route

1. Mahadev Kirana (₹8500)
2. Shree Store (₹5200)
3. Patel Mart (₹12000)

- 👉 Tap → open order
- 👉 Highlight next stop

3) ORDER EXECUTION (MOST CRITICAL)

Screen:

Retailer: Mahadev Kirana
Invoice: INV123

Items:

Sugar 10kg

Ordered: 10

Delivered: [10]

Oil 1L

Ordered: 5

Delivered: [4]

[Proceed to Payment]

Rules:

- Delivered $\leq$ Ordered
 - Auto-detect shortage
-

👉 This solves:

- Damage
 - Short delivery
-

 4) DELIVERY STATUS

Options:

- () Delivered [confirmed by in-app otp]
() Partially Delivered [confirmed by in-app otp]
() Not Delivered
-

If NOT delivered:

Reason:

- () Shop Closed
() Refused
() Other
-

👉 Mandatory reason

5) PAYMENT COLLECTION (ZERO ERROR ZONE)

Payment UI

Final Amount: ₹7800

Cash: [5000]

UPI: [2800] [take photo]

sign :[phase 2][take photo] when credit system is introduced

Remaining: ₹0 

[Confirm Payment]

Rules:

- Cash + UPI = Final amount
- No mismatch allowed

 This is your **core trust engine**

6) CONFIRM DELIVERY

Summary:

Delivered: ₹7800

Cash: ₹5000

UPI: ₹2800 [take photo proof] [dynamic upi]

[OTP in app]

[Complete Delivery]

 Locks the transaction

7) SUCCESS SCREEN

 Delivery Completed

Collected: ₹7800

[Next Stop]

 Fast transition

8) EXCEPTION HANDLING

Case 1: Partial Delivery

- Qty adjusted
 - Amount auto-reduced
-

Case 2: Shop Closed

- Mark & skip
-

Case 3: Damage

- Optional photo
-

 All tracked in the system

9) END OF DAY SETTLEMENT

Settlement UI

6

Day Summary

Orders: 15
Delivered: 13
Failed: 2

Cash: ₹60,000 (along with list of shop names)
UPI: ₹25,000 (along with list of shop names)

Expected: ₹85,000

Difference: ₹0 

[Close Day]

👉 Admin collects cash [completes settlement]

CRITICAL SYSTEM RULES

1. No payment → no completion
 2. No mismatch allowed
 3. No editing after submit
 4. Full audit log
-

REAL-WORLD OPTIMIZATION

1. Offline Mode (if possible)

- Save locally
 - Sync later
-

2. Big Buttons

- Drivers not tech-savvy
-

3. Minimal Typing

- Use dropdowns

4. Fast Screens

- <2 sec load
-

WHAT MAKES THIS SYSTEM POWERFUL

You control:

- 👉 Delivery
 - 👉 Payment
 - 👉 Accountability
-

👉 FINAL RESULT

Driver app becomes:

Execution engine + money control system

[phase 2]

GOAL

Reduce delivery time + fuel + chaos

Increase orders per driver

WHAT YOU ACTUALLY NEED (NOT COMPLEX AI)

INPUTS

Your system already has:

- Retailer location (lat/lng)
- Order size (₹ / weight proxy)
- Delivery priority
- Vehicle capacity

👉 That's enough

1) CLUSTERING (STEP 1 – MOST IMPORTANT)

Concept:

Group nearby shops together

Route Clustering Visualization

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Logic:

Group orders within 1–2 km radius

👉 **Result:**

- One driver → one area
 - No zig-zag movement
-

🔥 **2) SIMPLE ROUTE ORDERING (NO COMPLEX AI)**

Use:

👉 **Nearest Neighbor Logic**

Algorithm:

Start from warehouse

Pick nearest shop

Then next nearest

Then next nearest

👉 **Works 80% cases**

👉 **Very fast to implement**

🔥 **3) PRIORITY LAYER (SMART TWIST)**

🔥 **4) Vehicle CAPACITY LOGIC**

Don't overload:

Example:

Vehicle capacity = 2500 kg

Assign orders until limit reached

 **Avoid:**

- **Delivery delays**
 - **Re-routing**
-

 **5) TIME SLOT LOGIC (OPTIONAL)**

Example:

Restaurant → morning delivery

Kirana → afternoon

 **Improves success rate**

 **DRIVER UI (SMART ADDITION)**

Show:

Next Stop: Mahadev Kirana

Distance: 1.2 km

ETA: 5 mins

 **Simple guidance, not maps overload**

 **OPTIONAL UPGRADE (PHASE 3)**

Use Google Maps API

Google Maps Platform

For:

- Traffic-based ETA
- Better route optimisation

👉 Add later, not now

⚠️ WHAT NOT TO DO

-
- ❌ Complex AI models
 - ❌ Over-optimisation
 - ❌ Real-time recalculation
-

👉 Not needed at your stage

💡 REAL INSIGHT

In India, FMCG:

👉 Biggest inefficiency is:

- Bad grouping
 - Not route math
-

🚀 FINAL SYSTEM

Orders → Cluster → Assign Driver → Sort by Distance + Priority → Execute

👍 RESULT

You get:

- Faster delivery
- Lower fuel cost
- More orders per driver

Additional notes

- *Security Module*

- Implement Face ID authentication in all applications
- Implement the fingerprint lock system in all applications
- App lock functionality for enhanced security

- *Salesman Module*

- A salesman can edit the product price while placing retailer orders
- The distributor can define a maximum discount limit for the salesman
- Salesman QR scanner for retailer identification and order placement

- *Retailer Module*

- Retailers can create orders directly from the POS
- Retailers can edit product prices during order creation
- Retailer QR code generated after the salesman's approval
- The order can be rescheduled if the retailer is unavailable
- The order can be cancelled if the retailer's shop is closed/unavailable

- *Distributor Module*

- The distributor can manage maximum discount permissions for salesmen
- Subscription page in the distributor panel and app
- Display saved amount and billing overview for the distributor

- *Order Management Module*

- Order editing support for distributors and salesmen
- Order reschedule functionality
- Order cancellation functionality

- *Notification Module*

- Send notification to retailer when distributor edits order
- Send a notification to the retailer when a salesman edits order

- *QR Code Module*

- Retailer QR code generation after approval
- QR scanning support for the salesperson application
- Quick retailer identification using a QR system